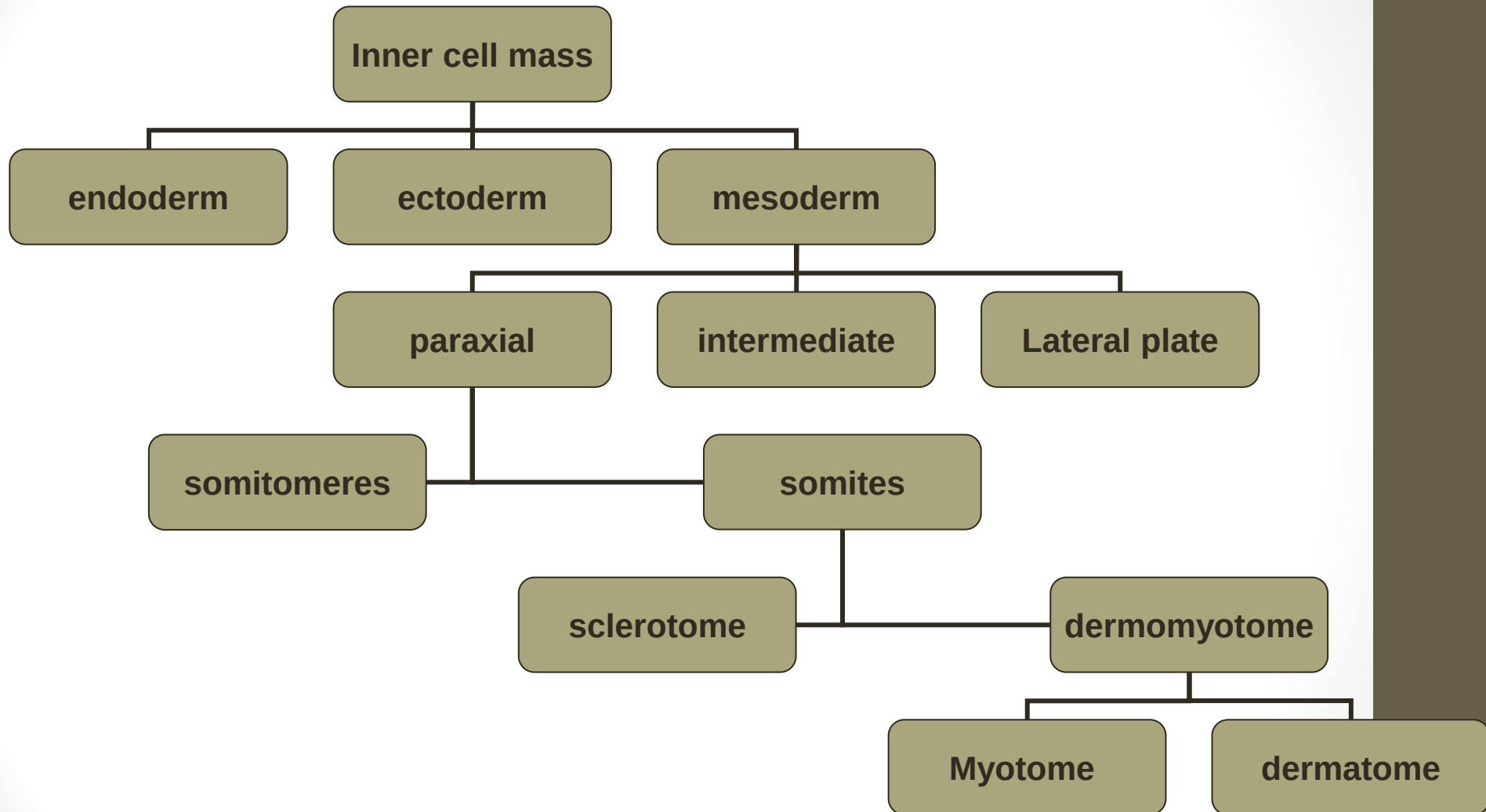


Development of skeletal system

Introduction

- The skeletal system develops from
 - **Paraxial mesoderm**
 - **Lateral plate (somatic layer) mesoderm**
 - **Neural crest**
- Paraxial mesoderm forms a segmented series of tissue blocks on each side of the neural tube, known as **occipital somitomeres** in the head region and **somites** caudal to the occipital region.
- Somites differentiate into a ventromedial part, the **sclerotome**, and a dorsolateral part, the **dermomyotome**

Origin and fate of paraxial mesoderm



Cont...

- At the end of the 4th week sclerotome cells become differentiated into **mesenchyme**
- Mesenchymal cells migrate and **differentiate** in many ways and become **fibroblasts, chondroblasts, or osteoblasts**
- In some bones, such as the flat bones of the skull, mesenchyme differentiates directly into **bone**, a process known as **intramembranous ossification**
- In most bones, however, mesenchymal cells first give rise to **hyaline cartilage models**, which in turn become ossified by a process known as **endochondral ossification**

Cont...

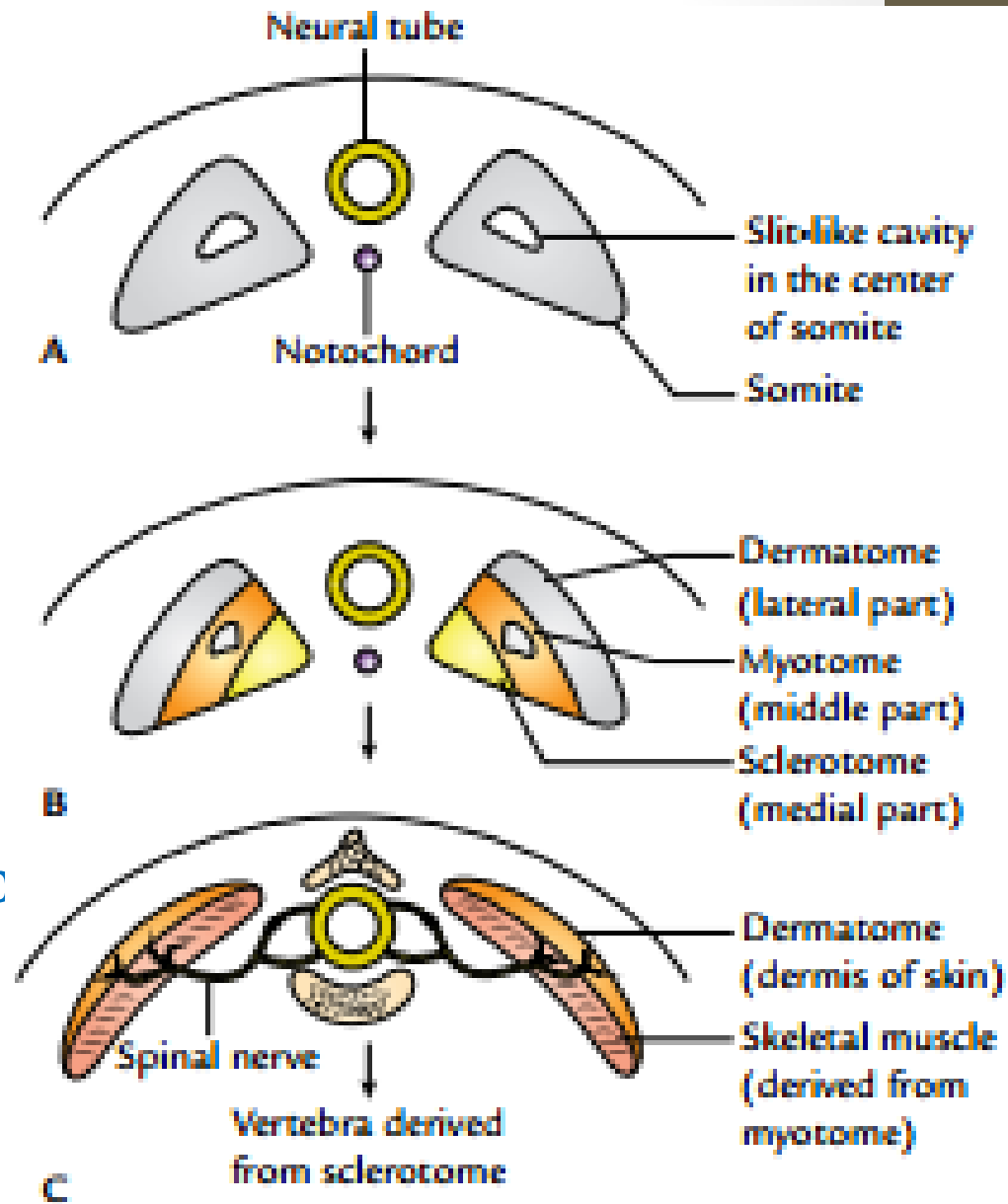
- The bone-forming capacity of mesenchyme is not restricted to cells of the sclerotome, but occurs also in the **somatic layer of mesoderm** also contributes for the formation of the **pelvic** and **shoulder girdles** and the bones of the **limbs**
- Neural crest cells in the head region also differentiate into mesenchyme and participate in formation of bones of the **face** and **skull**
- **Occipital somitomeres** also contribute to formation of the cranial vault and base of the skull

Development of the somite

- Somites are cubical blocks of paraxial mesoderm located each side of developing neural tube.
- The somites first appear in the future **occipital region** of the embryo.
- Subsequent pairs form in craniocaudal sequence.
- Cells in the **antromedial** wall lose their epithelial arrangement and become mesenchymal this collectively are called the **sclerotome**
- And a **dorsolateral** part called **dermomyotome**

Cont...

- The **sclerotomes** take part in the development of the **vertebral column**.
- Cells in the **dorsolateral** wall(myotome) of the somite form **musculature c** **limb** and **body wall**



Skeleton

- **Axial skeleton:** originate from **sclerotome**
- **Appendicular skeleton:** derived from **lateral plate mesoderm**
- **Head**
 - some cranial bones: from **paraxial mesodermal of occipital somatomers.**
 - **facial bones** and some part of vault: from **neural crest**

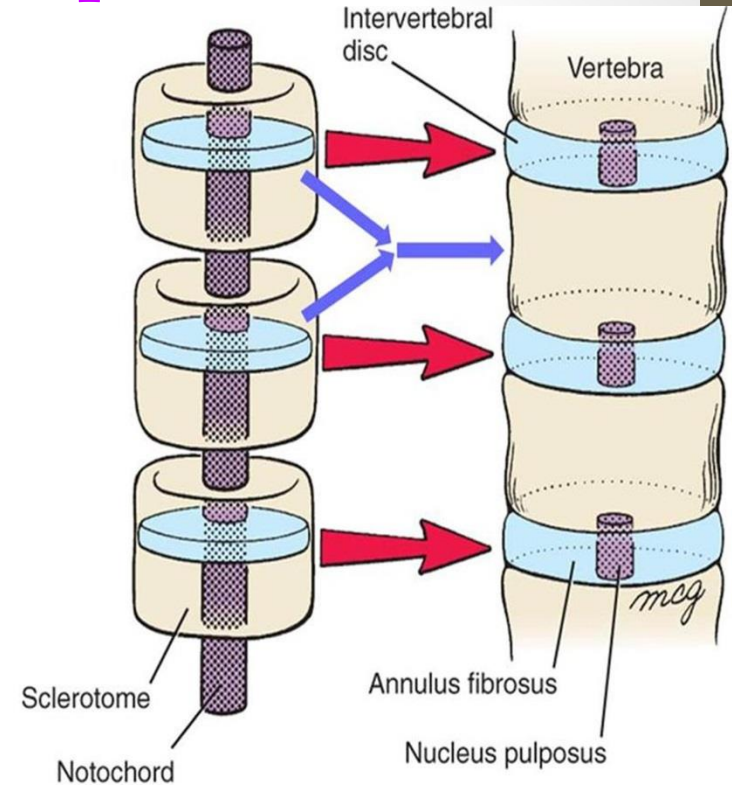
Axial skeleton

Vertebral column development

- During the fourth week of development, mesenchymal cells from the sclerotomes migrate **notochord** and **neural tube** to form the **vertebral column** as follows.
- The mesenchymal cells condense
 - a) around the notochord to form the **vertebral body**
 - b) around the neural tube to form right and left **vertebral arches**
 - c) in the body wall adjacent to the proximal part of neural arches to form **costal elements of the transverse processes**.
- The costal elements of transverse processes in the thoracic region form **ribs**

Vertebral column development...

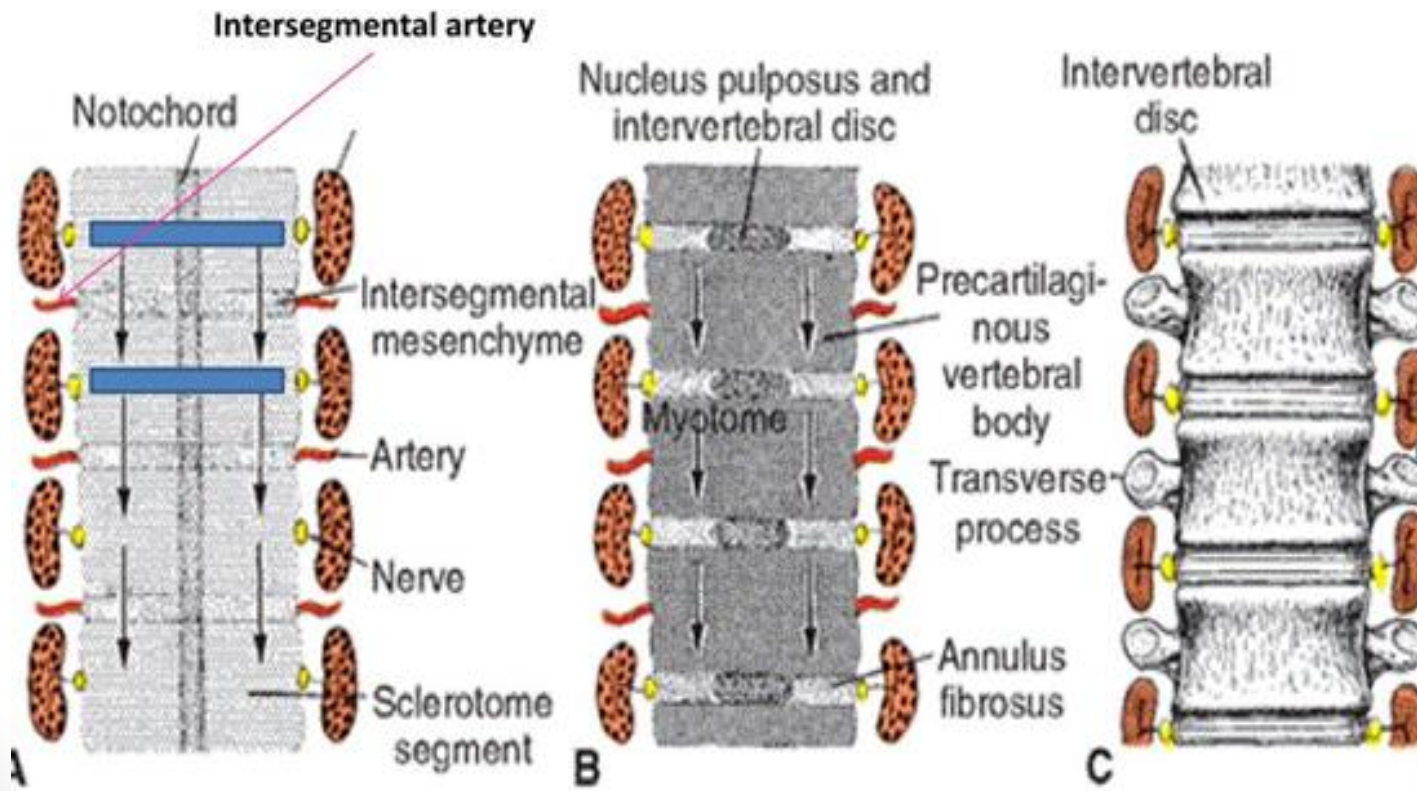
- As development continues, the sclerotome portion of each somite also undergoes **resegmentation**.
- **Resegmentation** occurs when the caudal half of each sclerotome grows into and fuses with the cephalic half of each subjacent sclerotome
- Thus, each vertebra is formed from the combination of the **caudal half** of one somite and the **cranial half** of its neighbor



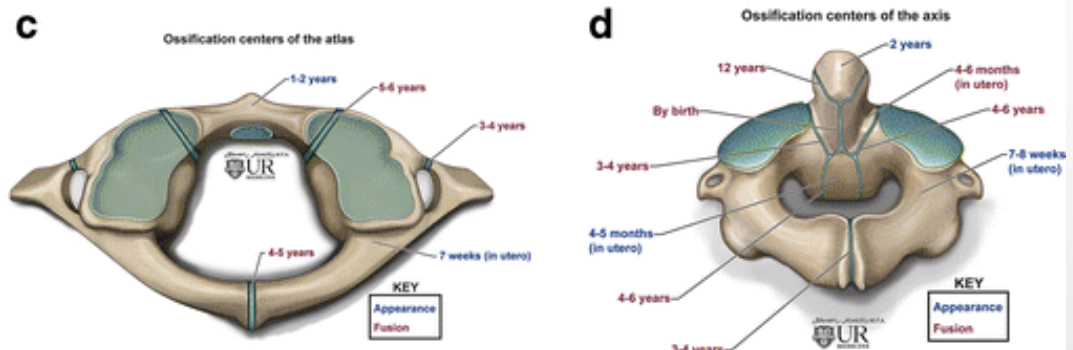
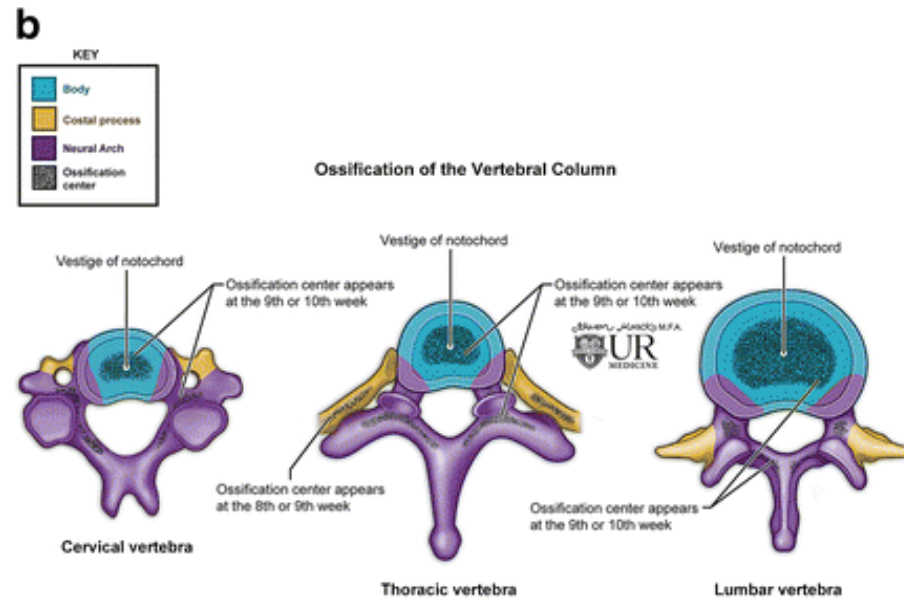
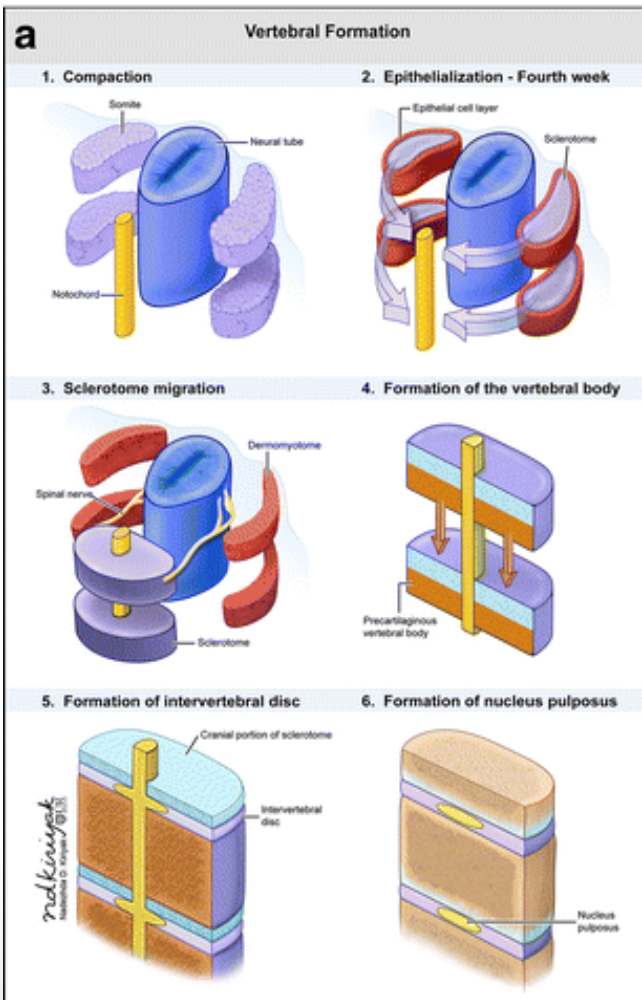
Vertebral column development...

- Mesenchymal cells between cephalic and caudal parts of the original sclerotome segment do not proliferate but fill the space between two precartilaginous vertebral bodies and contribute to formation of the **intervertebral disc**.
- Notochord **regress** in regions of **vertebral bodies** but **persists** and enlarges in regions of **intervertebral disc** and form **nucleus pulposus**

- Rearrangement of sclerotomes into definitive vertebrae causes
 - the myotomes to bridge the intervertebral discs
 - intersegmental arteries, can pass midway over the vertebral bodies
 - Spinal nerves, come to lie near the intervertebral discs and leave the vertebral column through the intervertebral foramina



Vertebral column development...



Congenital anomalies

Vertebral Defects

- When two successive vertebrae fuse asymmetrically or have half a vertebra missing, a cause of **scoliosis** (lateral curving of the spine]

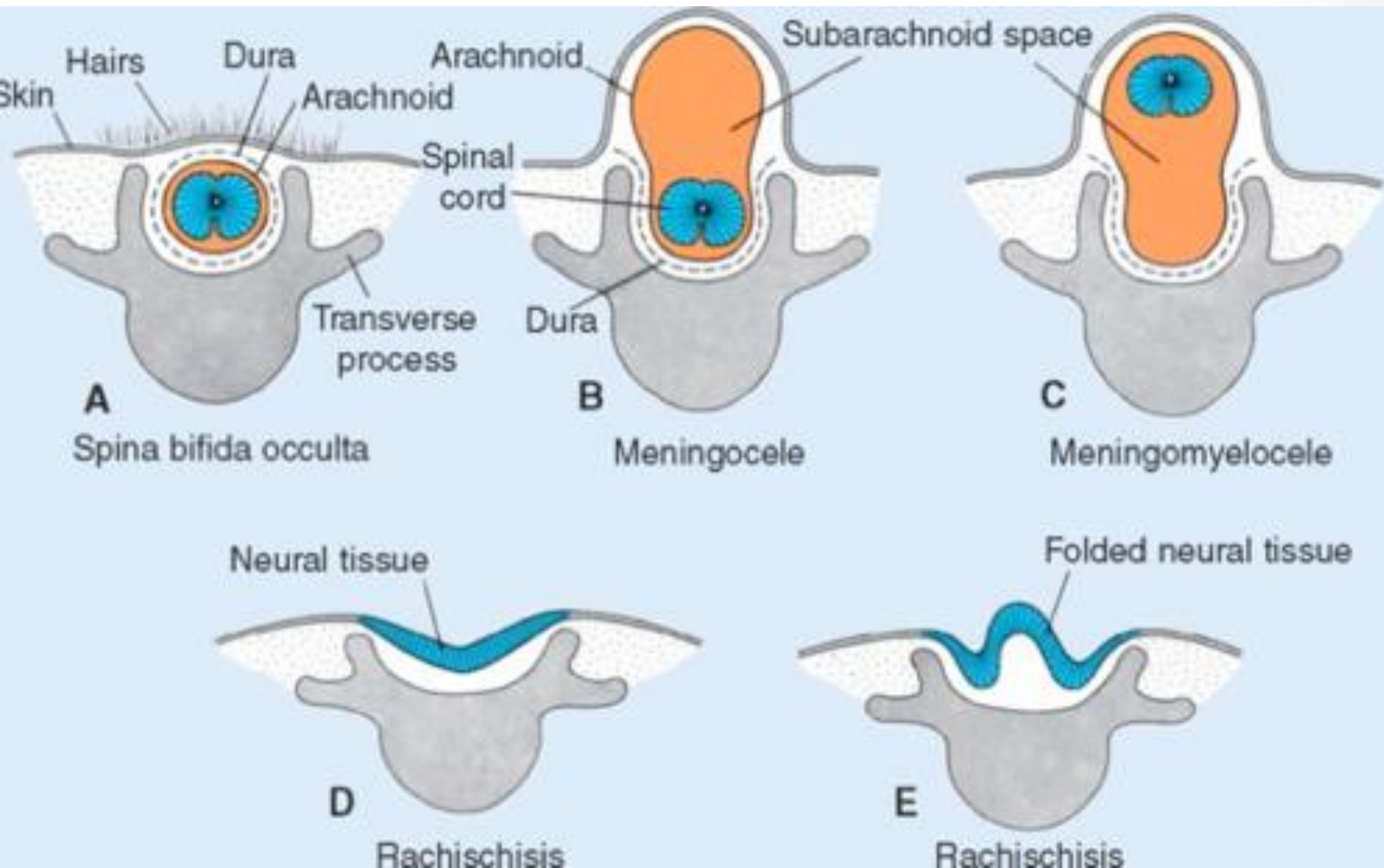
Spinal bifida - is imperfect fusion or nonunion of the vertebral arches.

A. Spina bifida occulta

B. Spina bifida cystica

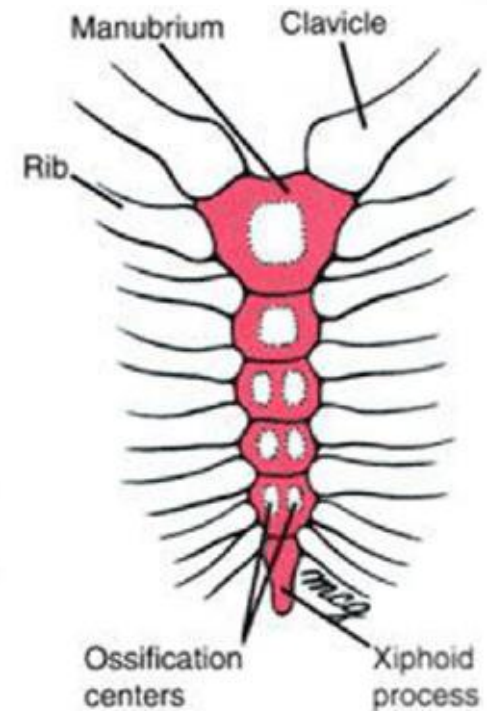
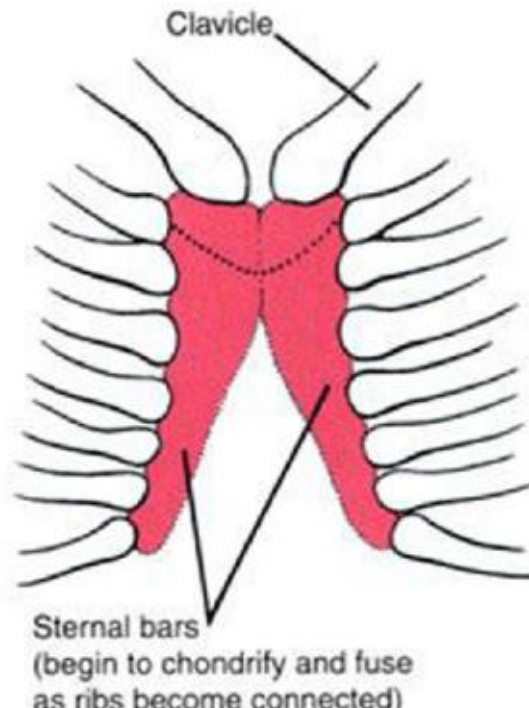
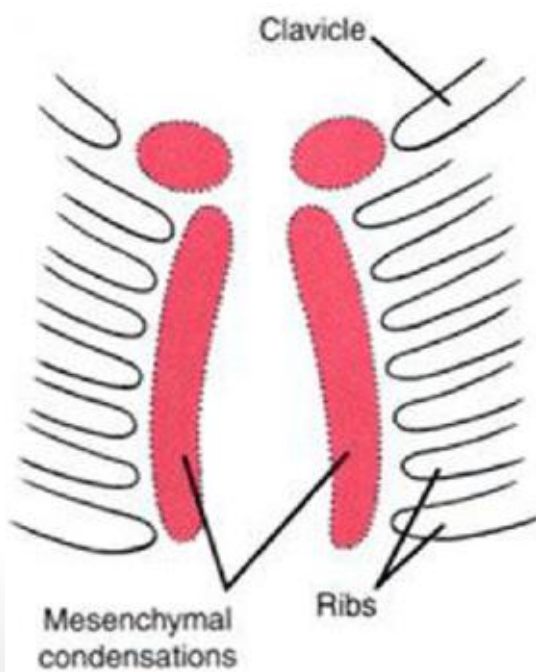
- I. Spina bifida with meningocele:
- II. Spina bifida with meningomyelocele:
- III. Spina bifida with rachischisis

Spina bifida



Sternum

- Develops independently in somatic mesoderm in the ventral body wall
- Two sternal bands are formed on either side of midline
- Fuse to form cartilage models of manubrium, body and xiphoid process
- Ossify craniocaudally before birth



Skull

- 2 parts:
 - **Neurocranium**: form protective case around **brain**
 - **Viscerocranium**: form skeleton of **face**

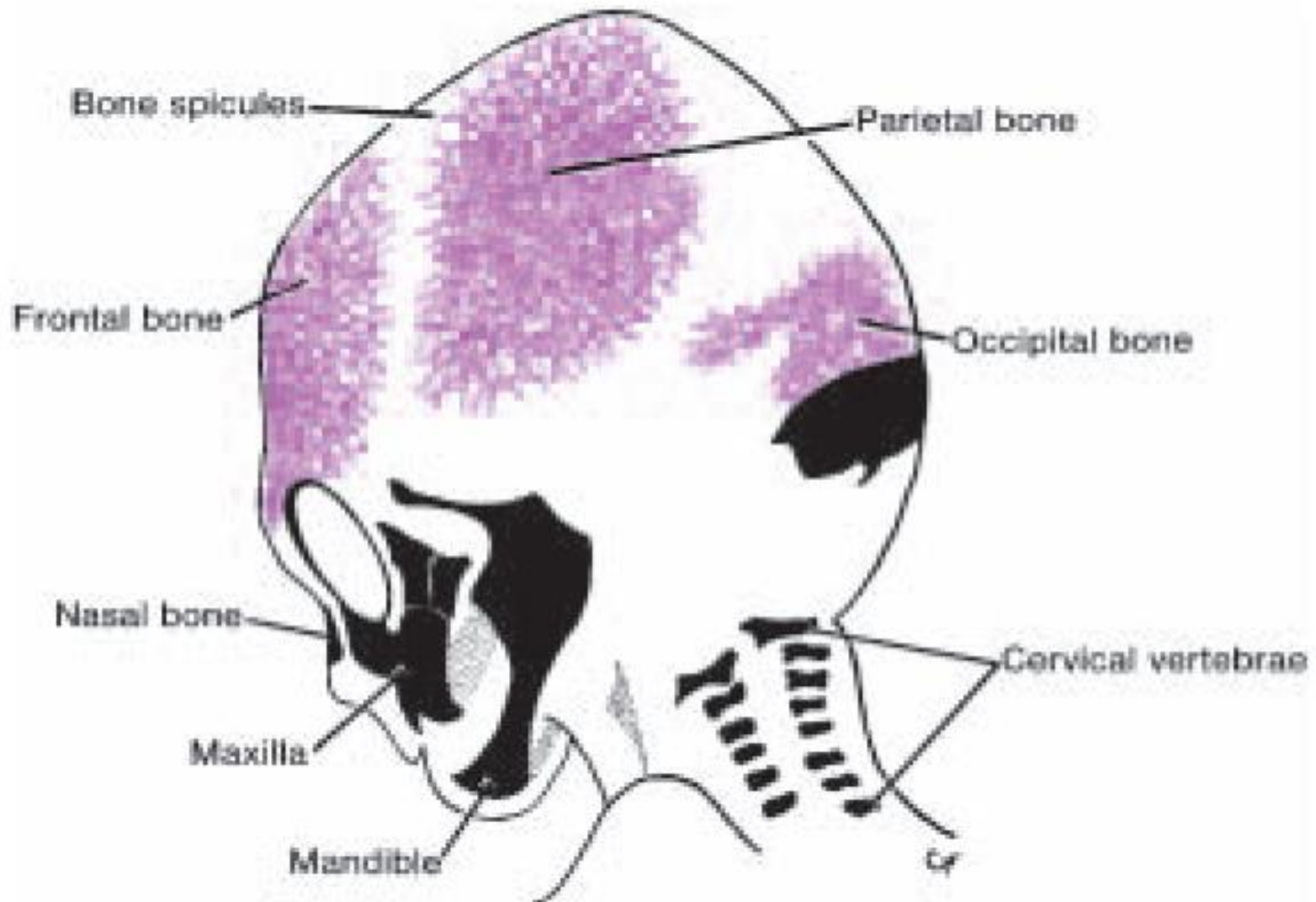
Neurocranium

- 2 portions
 - **Membranous part** - flat bones surround brain as vault
 - **Cartilaginous part (chondrocranium)** - forms bones of base of the skull

Membranous neurocranium

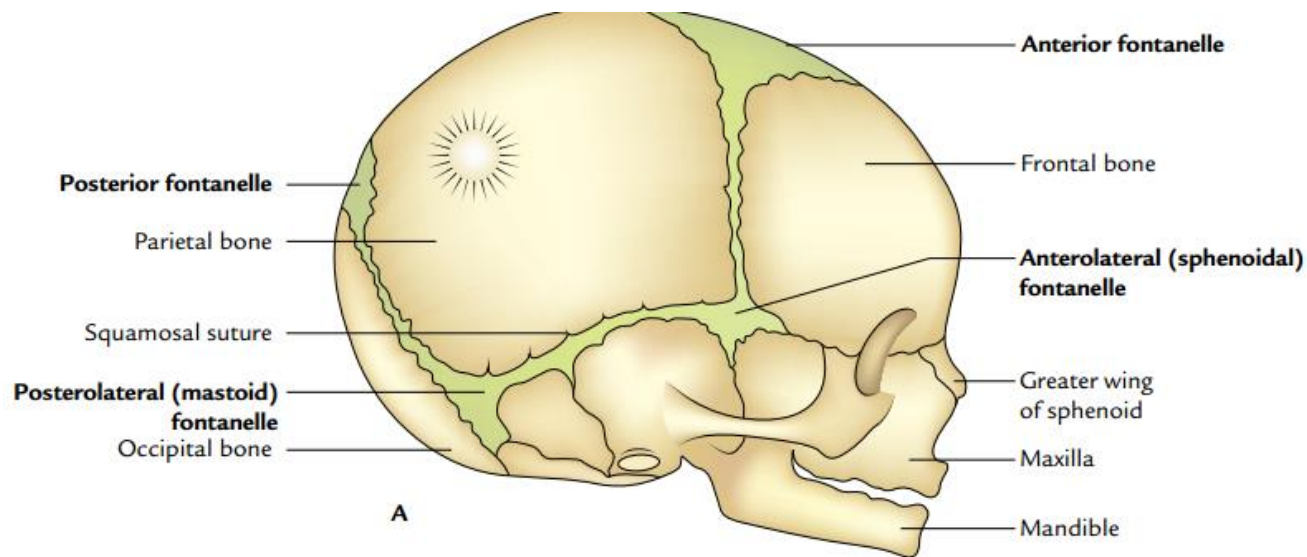
- The vault of skull develops from mesenchymal cells that derived from the **neural crest** cells and **paraxial mesoderm**.
- The vault of the skull is first membranous in nature that later undergoes **membranous ossification** to form a number of flat membranous bones, which together form the vault (top) of the skull.
- The primary centers appear where the bones start ossifying and grow towards the periphery by forming needle-like bony **spicules**.

Bones of the skull of a 3-month-old fetus showing the spread of bone spicules from primary ossification centers in the flat bones of the skull



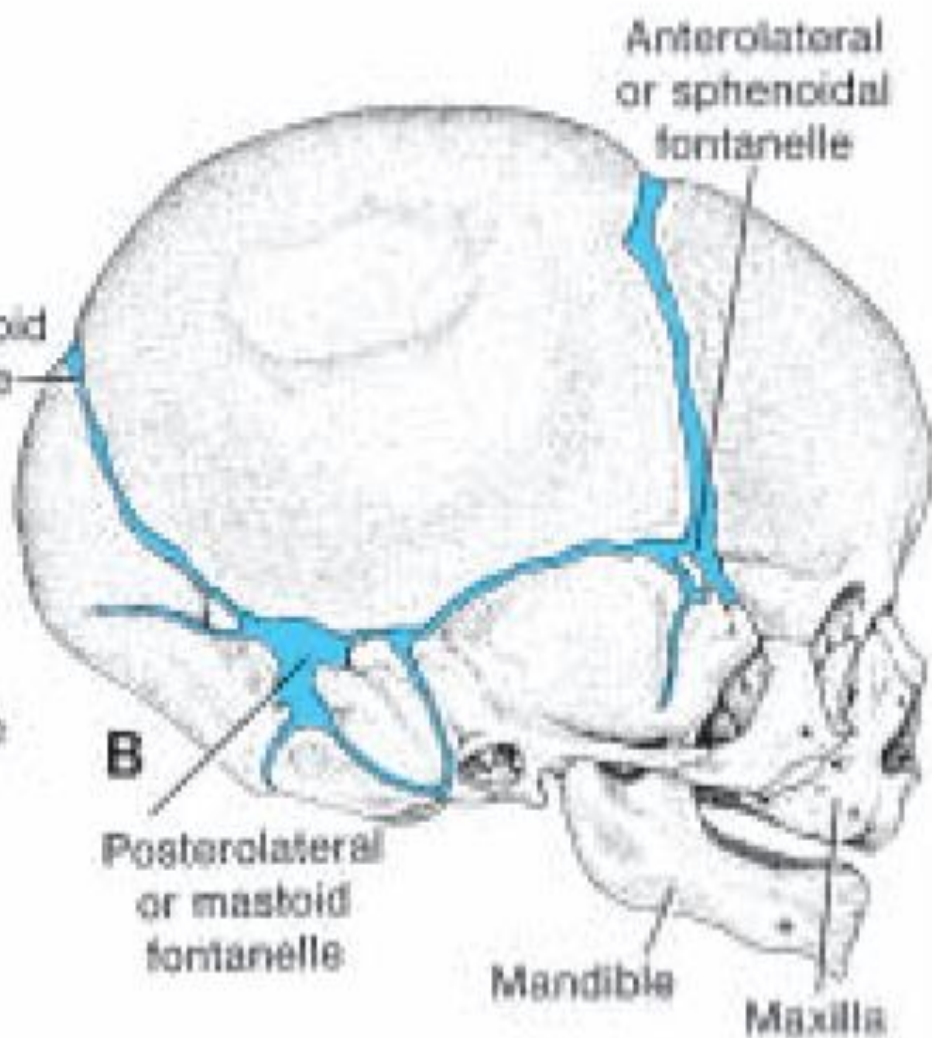
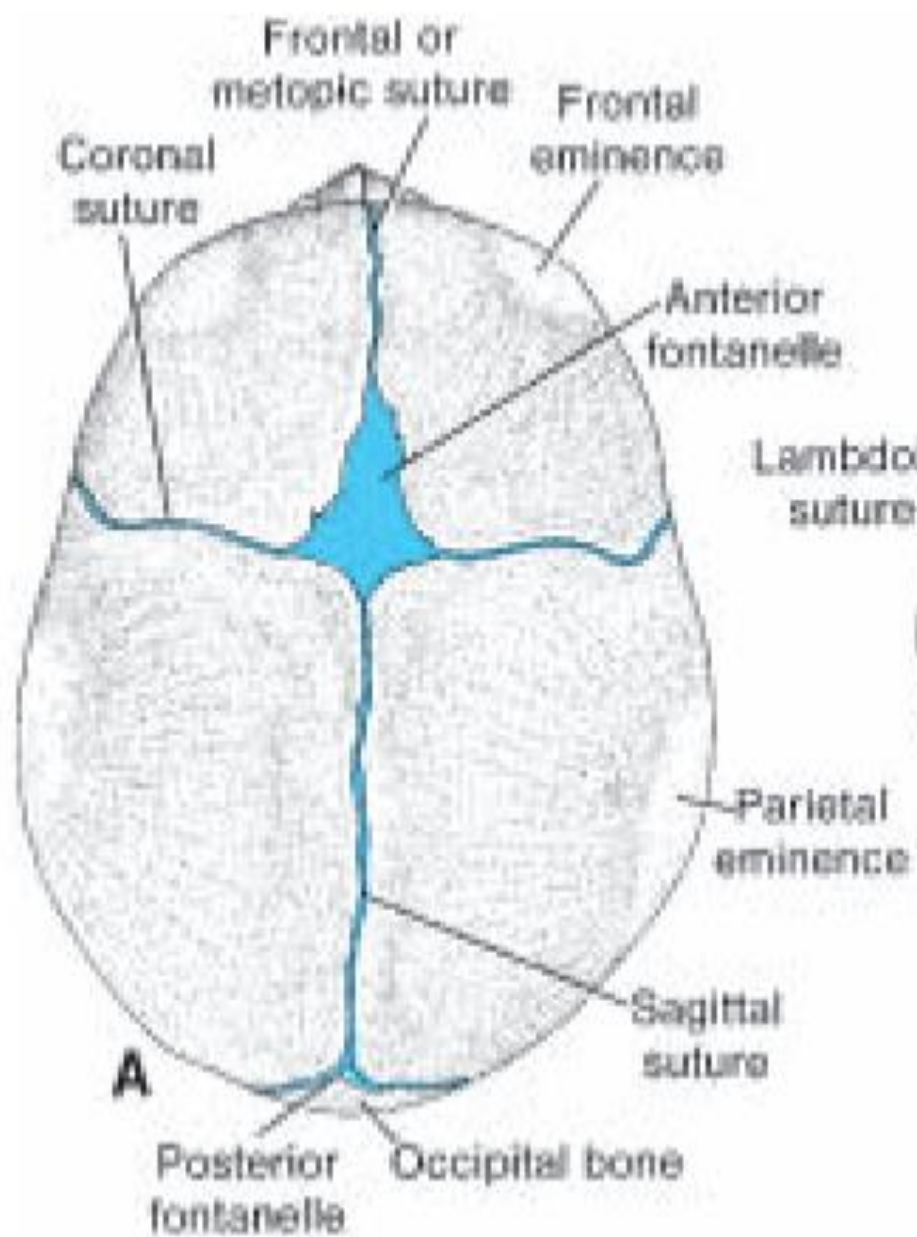
New born Skull

- At birth the flat bones of the skull are separated from each other by narrow seams of connective tissue, the **sutures**
- At points where more than two bones meet, sutures are wide and are called **fontanelles**
- There are **six fontanelles at birth**
 - Anterior and posterior,
 - Two anterolateral (or sphenoid)
 - Two posterolateral (or mastoid).
- The most prominent of these is anterior fontanelle



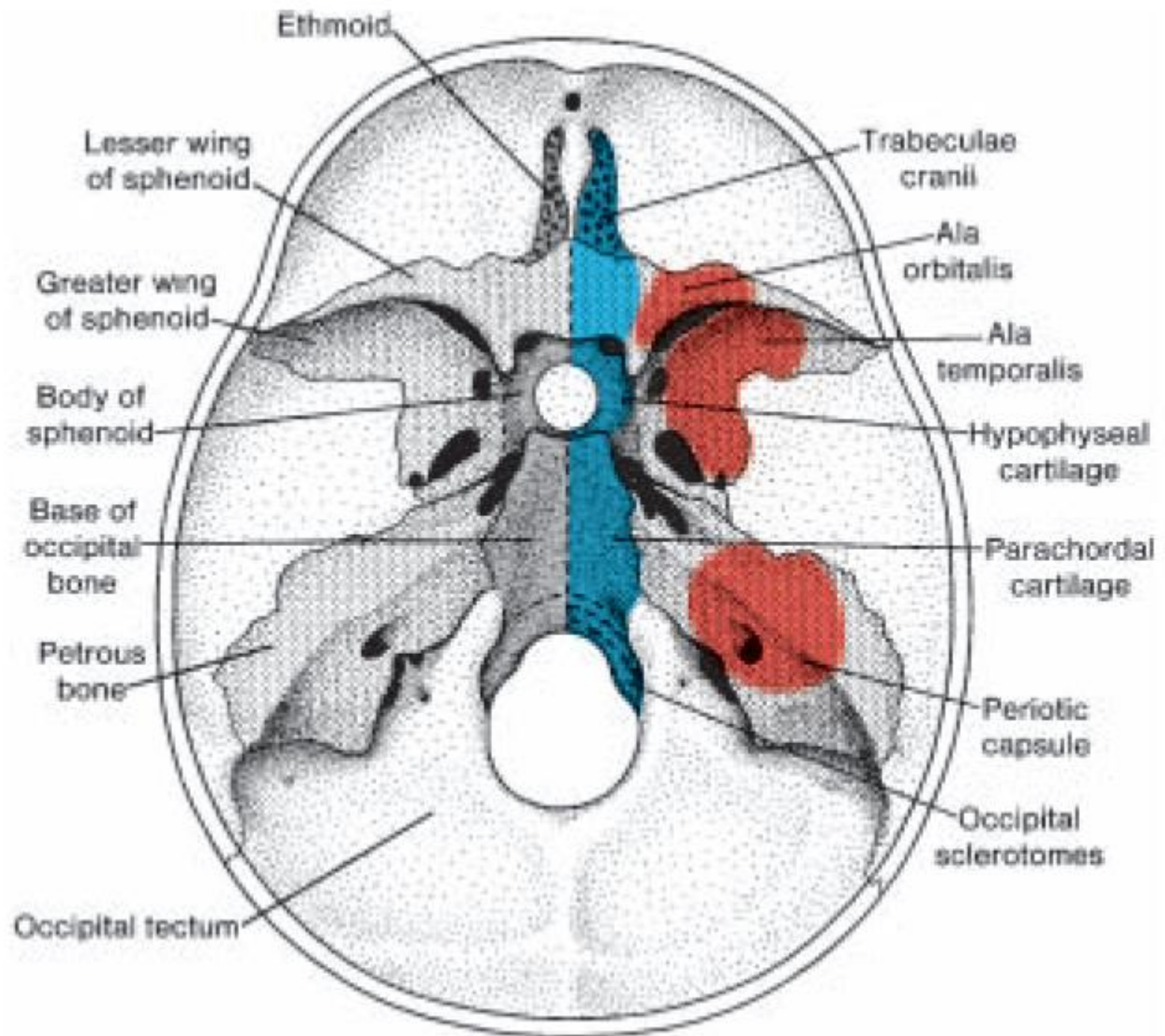
Cont...

- Sutures and fontanelles allow the bones of the skull to overlap (**molding**) during birth
- Most sutures and fontanelles remain membranous for a considerable time after birth
- The bones of the vault continue to grow after birth, mainly because the brain grows
- All the fontanelles except anterior fontanel closed within 3 months after birth
- The anterior fontanelle closed about the middle of the 2nd year
- Many of the sutures disappear during adult life



Chondrocranium (Cartilaginous Neurocranium)

- The base of skull, at first, is cartilaginous.
- It is formed by the fusion of several cartilages.
- These cartilages are formed by chondrification of mesenchyme below the brain.
- The base of the skull is formed when these **cartilages fuse** and ossify by **endochondral ossification**

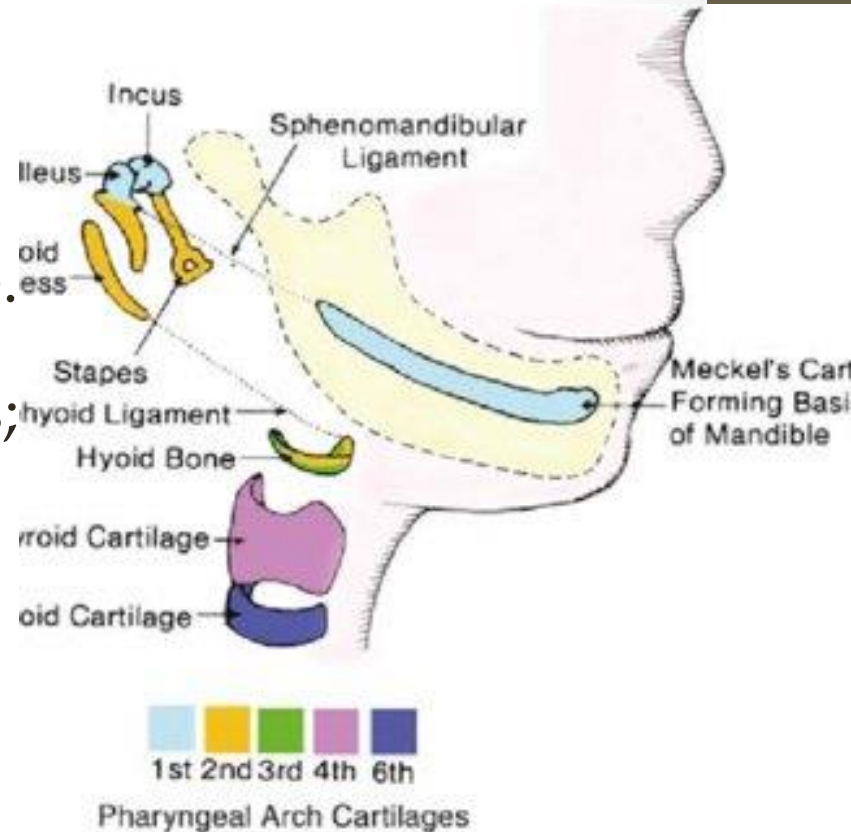


Viscerocranium

- Formed mainly from the first two pharyngeal arches
- 1st arch give rise to **dorsal (maxillary) and ventral (mandibular) processes**
- **Dorsal part** give rise to **maxilla, zygomatic** and part of **temporal bone**
- **Ventral part** contains **mandibular bone**
- The mandibular process of the first pharyngeal arch contains **Meckel's cartilage**.

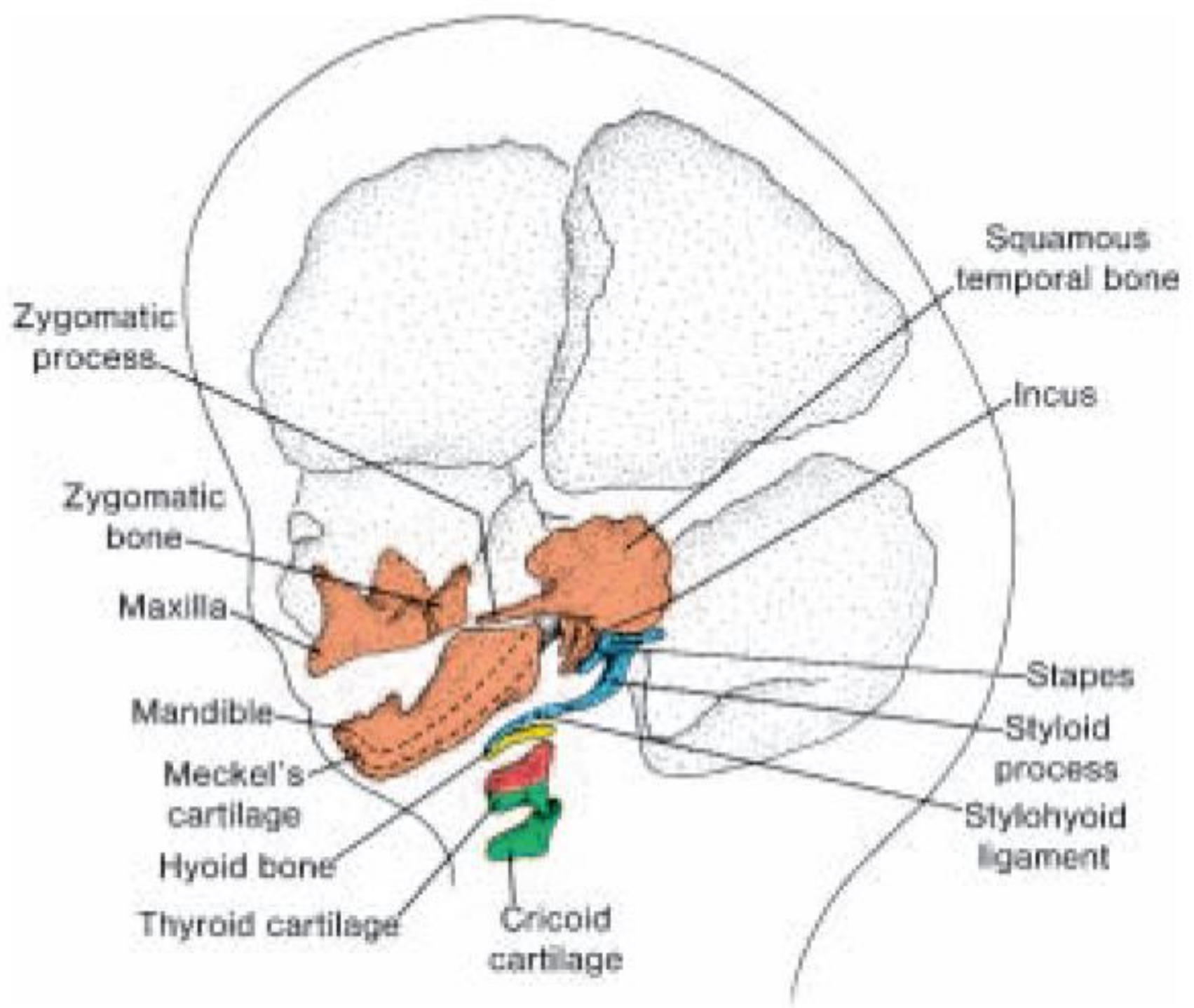
Cont....

- The mesenchyme around the Meckel's cartilage condenses and ossifies by membranous ossification to form the **mandible**.
- The Meckel's cartilage disappears; however its perichondrial sheath remains in the form of **sphenomandibular ligament**.



Cont....

- Mesenchyme for formation of the bones of the face is derived from **neural crest cells**, including the **nasal** and **lacrimal bones**
- At first the face is **small** in comparison with the neurocranium, this is caused by
 - absence of the paranasal air sinuses
 - the small size of the bones, particularly the jaws
- With the appearance of teeth and development of the air sinuses, the face loses its babyish characteristics



Developmental anomalies

Cranioschisis

- cranial vault fails to form due to failure of the **cranial neuropore to close**
- Brain tissue exposed to amniotic fluid and degenerate, resulting in **anencephaly**
- Children with such severe skull and brain defects cannot survive.
- **meningocele** or **meningoencephalocele**: brain tissue and/or meninges herniate through the defect in skull

• Craniosynostosis

- Caused by **premature closure** of one or more **sutures**
- The shape of the skull depends on which of the sutures closed prematurely
- Early closure of the **sagittal suture** results in frontal and occipital expansion, and the skull becomes long and narrow (**scaphocephaly**)
- Premature closure of the **coronal suture** results in a short, high skull, known as **acrocephaly**, or **oxycephaly** (tower skull)
- If the **coronal** and **lambdoid** sutures close prematurely on **one side only**, asymmetric craniosynostosis, known as **plagiocephaly**, results

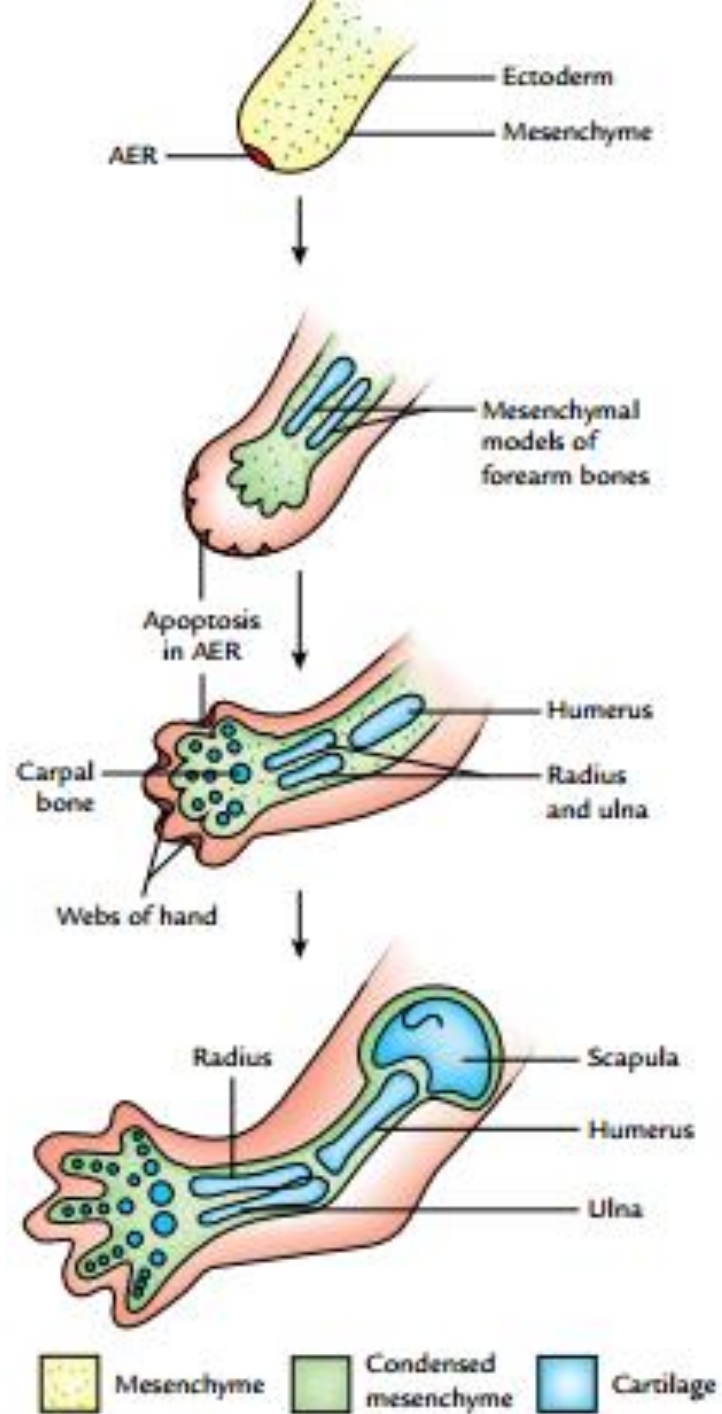
A. scaphocephaly **B.** acrocephaly **C.** plagiocephaly



- **Achondroplasia (ACH)**
- **Thanatophoric dysplasia**
- **Hypochondroplasia**
- **Acromegaly**
- **Microcephaly**

Development of Limbs

- The limb buds appear as out pouching's from the lateral aspects of the body wall at the end of the **4th week** of embryonic development.
- The buds of the upper limbs appear at around the **26th day** and that of the lower limb **1 or 2days later**.
- Initially, the limb buds consist o f a mesenchymal core derived from the **somatic layer of lateral plate mesoderm** that will form the bones and connective tissues of the limb, covered by ectoderm.
- The ectoderm at the tip of the limb bud thickens to form **apical ectodermal ridge (AER)**.



Development of Limbs...

- As the limb grows, cells farther from the influence of the AER begin to differentiate into **cartilage**.
- In this manner, development of each limb proceeds proximodistally into its three components:
- In the sixth week of embryonic development, terminal part of the limb bud becomes flattened to form **hand and foot plates**.

Development of Limbs...

- **Fingers** and **toes** are formed when cell death in the AER separates this ridge into five parts
- Now main parts of the limb become recognizable (e.g., **arm**, **forearm**, and **hand** in upper limb; and **thigh**, **leg**, and **foot** in lower limb).
- The development of upper and lower limbs is similar except following facts:
 - The development of lower limb starts 1 or 2 days later to that of the upper limb.
 - During the seventh week of gestation, upper limb rotates **90° laterally** whereas the lower limb rotates **90° medially**.